

TOWER OS

End-to-End TowerCo SaaS Platform

Board of Directors Presentation

Commercial SaaS Product for Tower Companies — Full Architecture & Roadmap

Classification: Confidential — Board Only

Version: 1.0 — Initial Architecture

Prepared by: Product & Engineering Team

Date: May 2026

1. Executive Summary

TowerOS is a comprehensive, cloud-native SaaS platform designed to serve tower companies (TowerCos) end-to-end — from site acquisition and project management through to asset tracking, field operations, fiber network management, and IoT site monitoring. It is a direct commercial alternative to enterprise platforms such as NexsysOne, built on a modern, scalable technology stack.

The platform is composed of 11 tightly integrated modules, each addressing a distinct operational domain. All modules share a single authentication layer, a centralized roles and permissions engine, and a multi-tenant PostgreSQL database — enabling TowerCos of any size to subscribe and operate from day one.

Strategic Objectives

- Build a commercially viable, multi-tenant SaaS product competing in the TowerCo software market
- Deliver all 11 operational modules on a unified platform with shared data and cross-module workflows
- Deploy on AWS cloud infrastructure for global availability, security, and scalability
- Generate recurring subscription revenue from tower companies worldwide
- Achieve first customer go-live within 12 months from project kick-off

Platform at a Glance

Modules: 11 fully integrated domain modules	Architecture: Multi-tenant SaaS — schema-per-tenant
Backend: Laravel 11 API (PHP)	Frontend: Next.js 14 (React, SSR + SPA)
Database: PostgreSQL 16 + PostGIS + TimescaleDB	Cloud: AWS (ECS Fargate, Aurora, ElastiCache)
Real-time: WebSocket via Laravel Echo + Socketi	Build timeline: 6 phases over ~18 months

2. Platform Modules

TowerOS is organized into 11 specialized modules. Each module is independently configurable per tenant but shares cross-module data (sites, users, assets, vendors) through a common data layer. The table below summarizes each module, its category, and its core feature set.

Module	Category	Core Features
PROJECT-ONE	Project & QMS	Quality management, project tracking, GIS visualization, PO/PR management, lease & contract management, documentation, build-to-suit, colocation
TOWER-ONE	Tower Management	Tower sharing & management, digital twin API, tenant billing, lease management, colocation optimization, site information hub
FIBER-ONE	Network / GIS	Fiber power routing, topology GIS tracking, connectivity & cross-connection management, transmission overlays, API integrations
ASSET-ONE	Asset & Logistics	Warehouse management, asset movement & RFID tracking, spare parts, supply chain, warranty, depreciation, logistics, barcode scanning
TASK-ONE	Field Workforce	Preventive & corrective maintenance, ticketing, SLA management, escalation, timesheet, live video communication, outage management, NOC integration
VENDOR-ONE	Vendor Management	Vendor onboarding, certification tracking, compliance, contract management, PO/invoice generation, supplier scorecards, shipment tracking
SITE-ONE	Site Monitoring	Satellite connectivity, energy monitoring, alarm handling, access control, IoT sensors, solar monitoring, generator management, fuel measurement
ADMIN-ONE	Platform Admin	Multi-tenant admin, roles & permissions, workflow builder, KPI/SLA config, milestone rules, report templates, bulk uploads, domain features
AUDIT-ONE	Quality & Auditing	Site surveys, audit processing, mobile app, automated acceptance, QMS templates, filterable audit reports, on-site quality checks
FLEET-ONE	Fleet Management	Real-time vehicle tracking, fuel monitoring, fleet maintenance, geofencing, trip management, work order routing, vehicle health alerts
ACCESS-ONE	Access Control	NOC access codes, schedule timing, access validation, remote access, smart lock integration, TASK-ONE workflow integration, field technician app

3. Technology Stack

TowerOS is built on a proven, production-grade open-source stack. The technology choices prioritize developer productivity, operational maturity, and the ability to support GIS features, real-time data, and IoT time-series workloads natively.

Backend — Laravel 11

- PHP 8.3 with Laravel 11 as the primary API framework
- RESTful JSON API with GraphQL support via Lighthouse PHP
- Authentication: Laravel Sanctum (SPA tokens) + Passport (OAuth2 for third-party integrations)
- Authorization: Spatie Laravel Permission — module-scoped roles and permissions
- Multi-tenancy: stancl/tenancy — schema-per-tenant PostgreSQL isolation
- Queues and background jobs: Laravel Horizon with Redis
- Real-time events: Laravel Echo + Soketi (self-hosted Pusher-compatible WebSocket server)
- Performance: Laravel Octane with Swoole for high-throughput API endpoints
- Observability: Laravel Telescope (development) + Sentry + Datadog APM (production)

Frontend — Next.js 14

- React 18 with Next.js 14 App Router — hybrid SSR and SPA rendering
- TypeScript for type safety across all frontend modules
- TailwindCSS + shadcn/ui component library for consistent design system
- React Query (TanStack) for server state management and cache synchronization
- Zustand for lightweight client-side state management
- Leaflet.js + MapLibre GL for GIS maps, fiber route overlays, and site mapping
- Recharts + Apache ECharts for dashboards, KPI charts, and analytics
- Socket.IO client for real-time NOC dashboard, live ticket updates, IoT feeds

Database Layer

Technology	Role	Reason
PostgreSQL 16	Primary database	ACID compliance, JSON support, robust multi-tenant schema isolation
PostGIS extension	GIS / geospatial	Native geospatial queries for FIBER-ONE, TOWER-ONE, and site mapping
TimescaleDB extension	IoT time-series	Efficient storage and querying of SITE-ONE sensor readings
Redis 7	Cache + queues	Session storage, API response caching, Laravel queue backend, pub/sub

Technology	Role	Reason
AWS S3	File storage	Document management, audit photos, firmware files, reports
Meilisearch	Full-text search	Fast, typo-tolerant search across assets, sites, vendors, and tickets

Cloud Infrastructure — AWS

- ECS Fargate: serverless container hosting — no EC2 instance management
- Aurora PostgreSQL: managed, auto-scaling database with multi-AZ failover
- ElastiCache: managed Redis for session, cache, and queue workloads
- API Gateway + CloudFront CDN: global edge delivery with WAF protection
- S3 + CloudFront: asset and document storage with global CDN
- SES + SNS: transactional email, SMS, and push notification delivery
- CodePipeline + ECR: fully automated CI/CD — GitHub push to production in minutes
- CloudWatch + X-Ray: infrastructure monitoring, distributed tracing, alerting

4. Core Database Schema

TowerOS uses a schema-per-tenant architecture on PostgreSQL. Each tenant company receives its own PostgreSQL schema (e.g., `tenant_acme`), while system tables (`tenants`, `global configs`) reside in the public schema. All primary keys use UUID v4. PostGIS geometry columns support GIS queries. TimescaleDB hypertables manage IoT time-series data.

Foundation Tables

tenants

Column	Type	Description
<code>id</code>	uuid PK	Unique tenant identifier
<code>name</code>	varchar	Company / tenant display name
<code>domain</code>	varchar	Custom subdomain (e.g. <code>acme.toweros.io</code>)
<code>schema_name</code>	varchar	PostgreSQL schema name for this tenant
<code>plan</code>	varchar	Subscription plan: starter, professional, enterprise
<code>is_active</code>	boolean	Whether the tenant account is active
<code>created_at</code>	timestamptz	Account creation timestamp

users

Column	Type	Description
<code>id</code>	uuid PK	Unique user identifier
<code>tenant_id</code>	uuid FK	Reference to <code>tenants.id</code>
<code>name</code>	varchar	Full name
<code>email</code>	varchar UNIQUE	Login email address
<code>password_hash</code>	varchar	Bcrypt hashed password
<code>mfa_enabled</code>	boolean	Whether MFA is active for this user
<code>last_login</code>	timestamptz	Timestamp of most recent login

sites

Column	Type	Description
<code>id</code>	uuid PK	Unique site identifier
<code>tenant_id</code>	uuid FK	Reference to <code>tenants.id</code>
<code>site_code</code>	varchar UNIQUE	Internal site code (e.g. <code>PH-MNL-001</code>)

Column	Type	Description
name	varchar	Site display name
latitude	decimal(10,8)	GPS latitude coordinate
longitude	decimal(11,8)	GPS longitude coordinate
geom	geometry(Point, 4326)	PostGIS geometry for spatial queries
type	varchar	Site type: macro, small_cell, rooftop, greenfield
status	varchar	Status: active, under_construction, decommissioned

PROJECT-ONE Tables

projects

Column	Type	Description
id	uuid PK	Unique project identifier
tenant_id	uuid FK	Reference to tenants.id
site_id	uuid FK	Reference to sites.id
name	varchar	Project name
project_manager_id	uuid FK	Reference to users.id
status	varchar	Status: planning, active, on_hold, completed, cancelled
start_date	date	Planned project start date
end_date	date	Planned project completion date

milestones

Column	Type	Description
id	uuid PK	Unique milestone identifier
project_id	uuid FK	Reference to projects.id
name	varchar	Milestone name (e.g. Site Acceptance, PAT)
due_date	date	Target completion date
status	varchar	Status: pending, in_progress, completed, overdue
order_index	integer	Sequence order within the project

purchase_orders

Column	Type	Description
id	uuid PK	Unique PO identifier
project_id	uuid FK	Reference to projects.id
vendor_id	uuid FK	Reference to vendors.id
po_number	varchar UNIQUE	Official PO reference number
amount	decimal(15,2)	Total PO value
currency	varchar(3)	ISO 4217 currency code
status	varchar	Status: draft, issued, received, cancelled

TOWER-ONE & ASSET-ONE Tables

towers

Column	Type	Description
id	uuid PK	Unique tower identifier
site_id	uuid FK	Reference to sites.id
tower_type	varchar	Type: guyed, monopole, lattice, rooftop
height_m	decimal(6,2)	Tower height in metres
capacity_kg	decimal(8,2)	Maximum structural load capacity in kg
max_tenants	integer	Maximum number of co-location tenants
status	varchar	Status: operational, under_maintenance, decommissioned

assets

Column	Type	Description
id	uuid PK	Unique asset identifier
tenant_id	uuid FK	Reference to tenants.id
asset_code	varchar UNIQUE	Internal asset code
name	varchar	Asset display name
category	varchar	Category: telecom_equipment, vehicle, tools, spare_parts
status	varchar	Status: in_warehouse, deployed, in_transit, under_repair, scrapped
rfid_tag	varchar	RFID tag identifier for hardware scanning
location_type	varchar	Location type: warehouse, site, vehicle
location_id	uuid	ID of the related warehouse, site, or vehicle

Column	Type	Description
warranty_expiry	date	Asset warranty expiry date
purchase_value	decimal(12,2)	Original purchase value

TASK-ONE & SITE-ONE Tables

work_orders

Column	Type	Description
id	uuid PK	Unique work order identifier
tenant_id	uuid FK	Reference to tenants.id
site_id	uuid FK	Reference to sites.id
assigned_to	uuid FK	Reference to users.id (field engineer)
wo_number	varchar UNIQUE	Official work order reference number
type	varchar	Type: preventive_maintenance, corrective, survey, installation
priority	varchar	Priority: critical, high, medium, low
status	varchar	Status: open, in_progress, on_hold, completed, cancelled
scheduled_at	timestamptz	Planned execution timestamp
sla_due_at	timestamptz	SLA deadline timestamp
completed_at	timestamptz	Actual completion timestamp

iot_devices

Column	Type	Description
id	uuid PK	Unique IoT device identifier
site_id	uuid FK	Reference to sites.id
device_type	varchar	Type: energy_meter, temperature, fuel_sensor, camera, access_lock
identifier	varchar	Hardware MAC address or device serial number
status	varchar	Status: online, offline, error
last_seen	timestamptz	Last successful data transmission timestamp

iot_readings (TimescaleDB hypertable)

Column	Type	Description
id	uuid PK	Reading identifier
device_id	uuid FK	Reference to iot_devices.id
metric	varchar	Metric name: voltage, current, temperature, fuel_level, door_state
value	decimal(12,4)	Numeric sensor reading value
unit	varchar	Measurement unit: V, A, C, L, bool
recorded_at	timestampz (partition key)	TimescaleDB partition column — time of reading

5. Project Phases & Roadmap

The TowerOS build is structured into 6 sequential phases over an estimated 18 months. Each phase delivers a working, deployable increment of the platform. Phases are ordered to ensure the foundational infrastructure and administration layer is complete before domain modules are built on top of it.

Phase	Duration	Modules	Key Deliverables	Milestone
Phase 1	Months 1–3	ADMIN-ONE	Multi-tenancy setup, auth system, role & permission engine, workflow builder, KPI/SLA configurator, onboarding portal, CI/CD pipeline	Platform foundation live
Phase 2	Months 4–6	PROJECT-ONE VENDOR-ONE	Project & milestone management, GIS site map, PO/PR workflow, document management, vendor onboarding, compliance tracking, supplier scorecards	First paying customer pilot
Phase 3	Months 7–9	TOWER-ONE ASSET-ONE	Tower sharing & co-location, tenant billing engine, digital twin API, warehouse management, RFID/barcode tracking, asset lifecycle, logistics module	Tower & asset operations live
Phase 4	Months 10–12	TASK-ONE AUDIT-ONE ACCESS-ONE	Field workforce management, ticketing & SLA, NOC dashboard, live video, site auditing mobile app, smart lock integration, access code system	Field operations live
Phase 5	Months 13–15	FIBER-ONE SITE-ONE	Fiber route GIS, cross-connection tracking, IoT device integration, energy monitoring, solar & generator management, CCTV & alarm handling	Network & IoT live
Phase 6	Months 16–18	FLEET-ONE + Platform	Vehicle tracking & geofencing, fuel monitoring, fleet maintenance, global analytics dashboard, white-label tenant portals, public API, marketplace launch	Full platform GA launch

Phase 1 — Foundation (Months 1–3)

This phase establishes the entire platform infrastructure. Nothing else can be built without it. The output is a working multi-tenant SaaS skeleton that any developer can build domain modules on top of.

- Set up AWS infrastructure: ECS Fargate, Aurora PostgreSQL, ElastiCache Redis, S3, CloudFront, CodePipeline
- Implement multi-tenancy using stancl/tenancy — schema-per-tenant with automated provisioning
- Build authentication: Laravel Sanctum (web) + Passport OAuth2 (API), MFA, SSO via SAML

- Implement RBAC engine: Spatie Permissions — module-level roles, permission groups, tenant-scoped
- Build ADMIN-ONE module: tenant onboarding, user management, workflow builder, KPI/SLA config, bulk imports
- Build the Next.js design system: component library, navigation shell, dashboard framework
- Set up CI/CD pipeline: GitHub Actions → ECR → ECS, automated testing, staging environment

Phase 2 — Core Operations (Months 4–6)

The first commercial modules. These two modules cover the workflows most TowerCos need daily and are the primary entry point for early adopters.

- PROJECT-ONE: project creation & lifecycle, milestone tracking, dependency management, GIS site map (Leaflet + PostGIS)
- PROJECT-ONE: PO/PR workflow with approval chains, lease & contract management, documentation management system
- PROJECT-ONE: reporting engine — milestone reports, progress dashboards, automated milestone notifications
- VENDOR-ONE: digital vendor onboarding flow, document & insurance tracking, compliance scoring
- VENDOR-ONE: contract milestone tracking, PO generation from PROJECT-ONE, invoice management, shipment tracking
- VENDOR-ONE: supplier performance scorecards, real-time dashboards, lead-time reporting
- First pilot customer onboarding — validate workflows against real operational data

Phase 3 — Tower Business & Assets (Months 7–9)

The commercial core of a TowerCo business: managing towers, co-location tenants, and physical assets.

- TOWER-ONE: tower inventory, structural data, tenant co-location management, capacity optimization
- TOWER-ONE: lease management lifecycle, monthly billing engine, automated invoice generation, payment tracking
- TOWER-ONE: digital twin API — publish tower data to external planning and RF tools
- ASSET-ONE: multi-location warehouse management — aisles, bays, bins, stock levels
- ASSET-ONE: RFID and barcode scanning integration (mobile scanner + web interface)
- ASSET-ONE: asset movement tracking — warehouse to site to vehicle, full chain of custody
- ASSET-ONE: spare parts management, reorder triggers, supply chain procurement, logistics coordination
- ASSET-ONE: warranty tracking, depreciation scheduling, scrap-to-revenue refurbishment workflow

Phase 4 — Field Operations (Months 10–12)

The modules that connect the back office to the field. This phase enables real-time field workforce management, quality auditing, and secure site access.

- TASK-ONE: work order lifecycle — creation, assignment, scheduling, field execution, closure
- TASK-ONE: trouble ticketing with severity levels, escalation rules, SLA countdown timers and alerts
- TASK-ONE: NOC dashboard — real-time ticket map, outage management, network incident tracking
- TASK-ONE: field engineer mobile app — task lists, forms, photo capture, live video to NOC (WebRTC)
- TASK-ONE: timesheet management, resource scheduling, KPI tracking vs targets
- AUDIT-ONE: mobile audit app — offline-capable, configurable QMS checklists, on-site photo evidence
- AUDIT-ONE: automated acceptance processing, payment trigger integration, audit history and reports
- ACCESS-ONE: NOC access code system, smart lock API integration, schedule-based access permissions
- ACCESS-ONE: TASK-ONE integration — work order triggers access code generation, field app deep link

Phase 5 — Network & IoT (Months 13–15)

The most technically complex phase — GIS fiber network management and live IoT monitoring.

- FIBER-ONE: fiber route GIS editor — draw routes on map, waypoints, distance calculation (PostGIS)
- FIBER-ONE: cross-connection and port pairing management, connectivity database, splice records
- FIBER-ONE: transmission overlay maps — microwave, cable, and electrical layers on a single GIS view
- FIBER-ONE: external integrations — REST API for network design tools and financial ERP systems
- SITE-ONE: IoT device registry — register, configure, and monitor all site sensors and controllers
- SITE-ONE: real-time sensor dashboard — energy, fuel, temperature, door status, generator state
- SITE-ONE: alarm handling — configurable alert rules, escalation to TASK-ONE tickets, NOC notification
- SITE-ONE: solar monitoring integration, generator maintenance scheduling, fuel consumption reports
- SITE-ONE: CCTV / perimeter security integration, movement detection alerts, access event logs

Phase 6 — Fleet & General Availability (Months 16–18)

The final phase completes the platform and prepares it for full commercial launch and marketplace distribution.

- FLEET-ONE: real-time GPS vehicle tracking map, geofencing around depots and project sites
- FLEET-ONE: fuel consumption monitoring, route optimization, trip history and reporting
- FLEET-ONE: fleet maintenance scheduling, roadworthiness alerts, vehicle fault integration with TASK-ONE
- Global analytics platform: cross-module executive dashboards, scheduled report delivery, data export API
- White-label portal: per-tenant custom branding, domain, color scheme, and logo
- Public REST API with developer documentation — enable third-party integrations and partner ecosystem
- Marketplace launch: subscription tier management, payment gateway integration, trial onboarding flow
- Security audit, penetration testing, SOC 2 Type I certification process initiated

6. Roles & Permissions Architecture

TowerOS uses a hierarchical, module-scoped role-based access control (RBAC) system built on the Spatie Laravel Permission package. Every permission is namespaced to a module, ensuring a user granted access to TASK-ONE cannot inadvertently access FIBER-ONE data.

System Roles (across all tenants)

- Super Admin — Anthropic/platform team. Full access to all tenants, modules, and system config.
- Tenant Admin — Customer administrator. Full access within their tenant. Manages users and roles.

Tenant-Defined Roles (examples)

Role	Typical Permissions
Project Manager	project-one.create, project-one.edit, project-one.report, vendor-one.view
Field Engineer	task-one.view, task-one.execute, audit-one.conduct, access-one.request
NOC Operator	task-one.assign, task-one.escalate, site-one.monitor, access-one.grant
Finance Manager	tower-one.billing.view, purchase_orders.approve, vendor-one.invoice
Warehouse Staff	asset-one.receive, asset-one.dispatch, asset-one.scan
Read-only Viewer	All modules .view only — no create, edit, or approve permissions

ADMIN-ONE allows tenant administrators to create unlimited custom roles and assign any combination of module permissions — without writing any code. Permission changes take effect immediately across all active sessions.

7. Security & Compliance

Authentication & Access Control

- All API endpoints protected by Laravel Sanctum token authentication
- OAuth2 via Laravel Passport for third-party integrations and partner API access
- Multi-factor authentication (TOTP) enforced for all admin and NOC user roles
- SSO support via SAML 2.0 for enterprise customers (Azure AD, Google Workspace, Okta)
- JWT tokens with configurable expiry — short-lived access tokens, rotating refresh tokens

Data Security

- Schema-per-tenant PostgreSQL isolation — tenant data is cryptographically separated at the database level
- All data encrypted at rest (AWS RDS AES-256) and in transit (TLS 1.3)
- AWS WAF protects API endpoints from injection attacks, DDoS, and bot abuse
- Complete audit trail — all create, update, delete operations are logged with user, timestamp, and IP
- PII fields (email, phone, GPS coordinates) can be encrypted at the column level per tenant configuration

Compliance Roadmap

- SOC 2 Type I: targeted 6 months after GA launch
- ISO 27001: targeted 12 months after GA launch
- GDPR compliance: built-in data retention policies, right-to-erasure API, consent management
- Penetration testing: independent third-party pen test prior to GA launch

8. Commercial & Revenue Model

Subscription Tiers

	Starter	Professional	Enterprise
Modules included	3 modules	Up to 7 modules	All 11 modules
Users	Up to 20	Up to 100	Unlimited
Sites	Up to 50	Up to 500	Unlimited
GIS features	Basic map view	Full GIS + routing	Full GIS + custom layers
API access	Not included	Read-only API	Full API + webhooks
SLA	Business hours	99.5% uptime SLA	99.9% uptime SLA
Support	Email support	Priority support	Dedicated CSM

Additional Revenue Streams

- Professional services: implementation, customization, data migration, training
- White-label licensing: larger TowerCos or system integrators can resell under their own brand
- API marketplace: third-party developers can build integrations and extensions
- Hardware bundles: partner with IoT sensor and RFID hardware vendors for bundled hardware + SaaS deals

9. Team & Resource Requirements

Engineering Team (Build Phase)

Role	Headcount	Responsibility
Laravel Backend Engineers	4	API development, module logic, queue workers, integrations
Next.js Frontend Engineers	3	UI components, dashboard, mobile-responsive layouts
GIS / PostGIS Specialist	1	FIBER-ONE, TOWER-ONE GIS layers, PostGIS queries
DevOps / AWS Engineer	1	Infrastructure, CI/CD, monitoring, security hardening
QA Engineer	2	Test automation, API testing, regression, UAT support
UI/UX Designer	1	Design system, user flows, prototyping, usability testing
Product Manager	1	Roadmap ownership, customer feedback, sprint planning
Tech Lead / Architect	1	Architecture decisions, code review, cross-module design

Total core team: 14 people. Additional subject matter experts (tower operations, fiber network, IoT) engaged as consultants during Phase 2 through Phase 5 to validate domain-specific workflows against real-world TowerCo operations.

10. Risk Register

Risk	Severity	Mitigation
Scope creep across 11 modules	High	Strict phase gating — each phase locked before next begins. MVP feature set defined per module before development starts.
GIS complexity (PostGIS, fiber routing)	High	Hire or contract a dedicated GIS specialist from Phase 1. Run PostGIS spikes in Phase 1 to de-risk before FIBER-ONE build.
IoT device heterogeneity (SITE-ONE)	Medium	Define a standardized MQTT/REST device adapter layer. Partners certify devices against the TowerOS device SDK.
Multi-tenancy data isolation failure	Critical	Automated schema isolation tests in CI pipeline. Third-party security audit of tenancy layer before any paying customer.
Key engineer attrition	Medium	Document all architecture decisions. Cross-train engineers across modules. Avoid single points of knowledge.
Market competition from incumbents	Medium	Fast delivery of first 3 modules as differentiator. Focus on UX quality and mobile-first field tools as key advantages.
AWS cost overrun	Low	Use AWS Cost Explorer from day one. Set billing alerts. Fargate auto-scaling with maximum limits per tenant plan.

11. Recommended Next Steps

Based on this architecture review, the following actions are recommended for board approval and immediate execution:

Board approval of project scope, budget, and 18-month roadmap

Recruit Tech Lead/Architect and DevOps Engineer — these are the two blocking hires for Phase 1

Establish AWS organization account, set up billing alerts, and provision staging environment

Identify and contract 1–2 pilot TowerCo customers to validate Phase 2 workflows

Commission UX research sprint: shadow field engineers and NOC operators at pilot customers

Define MVP feature list per module and lock Phase 1 sprint backlog

TowerOS — Built to power the next generation of tower companies.

Confidential — For Board Review Only — May 2026